BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8477

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. If an iron core coil of reactance 628Ω is placed in series with 450Ω resistance in AC circuit. The phase difference will be:

(A) 51.5°
(B) 60°
(C) 30°
(D) 45°

Q2. Energy band theory based on:

(A) Wave mechanical model
(B) Bohr atomic model
(C) Pauli exclusion principle
(D) Electronic configuration of electrons

Q3. In transistor with common emitter configuration, output voltage is at phase difference of:

(A) 90°
(B) 100°
(C) 120°
(D) 180°

Q4. An electronic computer is vast arrangement of electronic switches which are made from:

(A) Resistors
(B) Inductors
(C) Capacitors
(D) Transistors

Q5. h/m_oc has the unit of:

(A) Time
(B) Distance
(C) Velocity
(D) Acceleration

Q6. An electron moving with speed of $1\times10^{6}ms^{-1}$ has wavelength:

(A) $7\times10^{-10}m$
(B) $7\times10^{-9}m$
(C) $7\times10^{10}m$
(D) $7\times10^{-8}m$

Q7. Velocity of electron of hydrogen in different orbits is:

(A) Same
(B) Quantized
(C) Increase in higher orbits
(D) Independent of orbit number

Q8. It is believed that quark cannot exist in:

(A) Free state
(B) Bound state
(C) Quark, antiquark combination
(D) Three quark combination

Q9. Ultra violet radiation cause:

(A) Healthy growth
(B) Saver crop damage
(C) Fast hair grow
(D) Formation of ozone

Q10. Joule per Coulomb is equal to:

(A) Second
(B) Newton
(C) Watt
(D) Volt

Q11. Gravitational force cannot be:

(A) Mass dependent
(B) Distance dependent
(C) Shielded
(D) Stronger than electric force

Q12. In carbon resistance, colour bands are red, red, red and silver. The numerical value of resistance will be:

(A) 2200Ω±10%
(B) 220Ω±5%
(C) 22000Ω±20%
(D) 22Ω±10%

Q13. The torque on a current carrying rectangular coil placed outside the magnetic field will be:

(A) Maximum
(B) NIAB
(C) Zero
(D) IA cos θ

Q14. Sensitivity of moving coil galvanometer can be increased by:

(A) Decreasing area of coil
(B) Decreasing number of turns
(C) Using thick suspension wire
(D) Increasing magnetic field

Q15. When motor is just started, the current passing through the coil will be:

(A) Large
(B) Small
(C) Zero
(D) Average

Q16. The windings of electromagnet in DC motor are called:

(A) Solenoids
(B) Field coils
(C) Inductors
(D) Loops

Q17. When 10V is applied to an AC circuit with current of 10mA then impedance will be:

(A) 100Ω
(B) 10Ω
(C) 1000Ω
(D) 0.1Ω

Physics (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8477

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) If a point charge 'q' of mass 'm' is released in a non-uniform electric field with electric field lines pointing in the same direction, will it make a recti linear motion?

(ii) Describe the force of forces on a positive point charge when placed between parallel plates (i) with similar and equal charges (ii) with opposite and equal charges.

(iii) What is the potential gradient? Write its unit.

(iv) What is EEG and ERG?

(v) How can you use a current loop to determine the presence of a magnetic field in a given region of space?

(vi) Why the resistance of an ammeter should be very low?

(vii) Why parallel current attract and opposite current repel?

(viii) Distinguish between sensitive and dead-beat galvanometer.

(ix) How can radioactivity help in the treatment of Cancer?

(x) What do we mean by the term critical mass?

(xi) If $^{238}_{92}U$ decays twice by α-emission, what is the resulting isotope?

(xii) Write two advantages of solid-state detector?

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Write the heating effect of the current.

(ii) Why does the resistance of a conductor rise with temperature?

(iii) Explain why the terminal potential difference of a battery decreases when the current drawn from it is increased

(iv) What is a choke?

(v) How does doubling the frequency affect the reactance of a capacitor?

(vi) A sinusoidal current has rms value of 10A. What is the maximum or peak value?

(vii) Distinguish between hard magnetic material and soft magnetic material.

(viii) Define the terms yield point and ultimate tensile stress.

(ix) How the hysteresis loss is used in the construction of a transformer?

(x) Why is the base current in a transistor very small?

(xi) Write the truth table and Boolean expression of NAND gate.

(xii) What is the biasing requirement of the junctions of a transistor for its normal operation?

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Show that ε and $\Delta\varphi/\Delta t$ have the same units.

(ii) Can an electric motor be used to drive an electric generator with the output from the generator being used to operate the motor?

(iii) State Lenz's law. Does it agree with the law of conservation of energy?

(iv) Define mutual inductance and also define its unit.

(v) Is it possible to create a single electron from energy? Explain.

(vi) We do not notice de-Broglie wavelength for a pitched cricket ball. Explain why?

(vii) If the speed of light were infinite what would be the equations of special theory of relativity reduced?

(viii) Calculate the longest wavelength of radiation for Paschen series.

(ix) Is energy conserved when an atom emits a photon of light?

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 05) What is Wheatstone Bridge? How it can be used to find the unknown resistance?

Q5(b). (Marks 03) Determine the electric field at the position $\vec{r} = (4\hat{i} + 3\hat{j})$m caused by a point charge $q = 5.0\times10^{-6}$C placed at origin.

Q6(a). (Marks 05) State and explain Faraday's Law of electromagnetic induction.

Q6(b). (Marks 03) A power line 10m high carries a current 200A. Find the magnetic field of wire at the ground.

Q7(a). (Marks 05) Derive an expression for the resonance frequency in R.L.C series circuit. Also give properties of series resonance circuit.

Q7(b). (Marks 03) The current flowing into the base of a transistor is 100μA. Find its collector current Ic, its emitter current I_E and the ratio I_C/I_E; if the value of current gain β is 100.

Q8(a). (Marks 05) Explain energy band theory of solids. How does it help to distinguish between conductors, insulators and semi-conductors?

Q8(b). (Marks 03) What is the de-Broglie wavelength of an electron whose Kinetic energy is 120eV ?

Q9(a). (Marks 05) Write three postulate of Bohr's atomic model. Derive an expression for radii of quantized orbit of hydrogen atom?

Q9(b). (Marks 03) Find the energy associated with the following reaction: $^{21}_{7}N + ^{4}_{2}He \rightarrow ^{17}_{8}O + ^{1}_{1}H$. What does negative sign indicate?

GROUP – II

Physics (Objective) Group - II

Time Allowed: 20 Minutes	**Maximum Marks:** 17	**Paper Code:** 8474

> Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The energy released by fusion of two deuterons into a helium nucleus is about:

(A) 24 MeV
(B) 200 MeV
(C) 1.02 MeV
(D) 7.2 MeV

Q2. One Joule of energy absorbed in a body per kg is equal to:

(A) 1 rad
(B) 1 rem
(C) Sievert
(D) Gray

Q3. Paschan series is obtained when all the transitions of electron terminate on:

(A) 2nd orbit
(B) 3rd orbit
(C) 4th orbit
(D) 5th orbit

Q4. Platinum wire becomes yellow at room temperature of:

(A) 900°C
(B) 1300°C
(C) 1600°C
(D) 500°C

Q5. If object moves with the speed of light, its mass become:

(A) Zero
(B) Small
(C) Same
(D) Infinity

Q6. The thickness of a base in a transistor is of the order of:

(A) 10^{-3}m
(B) 10^{-4}m
(C) 10^{-6}m
(D) 10^{2}m

Q7. $X = \bar{A}\cdot\bar{B}$ is the mathematical notation for:

(A) NAND gate
(B) NOR gate
(C) OR gate
(D) AND gate

Q8. The critical temperature for mercury is:

(A) 7.2K
(B) 4.2K
(C) 1.18K
(D) 3.7K

Q9. Resistance of choke is:

(A) Zero
(B) Very small
(C) Large
(D) Infinite

Q10. In three phase, voltage across any two lines is:

(A) 220V
(B) 230V
(C) 400V
(D) 430V

Q11. In DC generator, split rings acts as:

(A) Capacitor
(B) Commutator
(C) Inductor
(D) Resistor

Q12. Energy stored in the inductor is:

(A) $\frac{1}{2}L^2I$
(B) $\frac{1}{2}LI$
(C) $\frac{1}{2}LI^2$
(D) $\frac{1}{2}L^2I^2$

Q13. A galvanometer becomes more sensitive when the factor C/BAN will be:

(A) Large (B) Small

(C) Constant (D) Zero

Q14. Force on a moving charge in a uniform magnetic field will be maximum, when the angle between $\vec{V}$ and $\vec{B}$ is:

(A) 0° (B) 30°

(C) 60° (D) 90°

Q15. Kirchhoff's first rule is based on conservation of:

(A) Energy (B) Voltage

(C) Charge (D) Mass

Q16. Coulomb per volt is called:

(A) Farad (B) Ampere

(C) Joule (D) Ohm

Q17. If the distance between two point charges is halved, the electric force becomes:

(A) Half (B) Double

(C) ¼ times (D) 4 times

Physics (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8474

> Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Is E necessarily zero inside a charged rubber balloon if balloon is spherical? Assume that charge is distributed uniformly over the surface.

(ii) Describe the force or forces on a positive point charge when placed between parallel plates (a) with similar and equal charges (b) with opposite and equal charges.

(iii) Derive relation for potential gradient $E = -\Delta V/\Delta r$

(iv) Write similarities and differences between electrostatic and gravitational forces.

(v) Define CRO and write the names of its parts.

(vi) Define tesla and write its formula.

(vii) Why does the picture on a TV screen become distorted when a magnet is brought near the screen?

(viii) What should be the orientation of a current carrying coil in a magnetic field so that the torque acting upon the coil is (a) maximum (b) minimum?

(ix) What do you understand by "back ground radiation"? State two sources of this radiation.

(x) What do you mean by the term critical mass?

(xi) Define absorbed dose and gray.

(xii) Write the names of basic forces of nature.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Is the filament resistance lower or higher in a 500W, 220V light bulb than in a 100W, 220V bulb?

(ii) Why does the resistance of conductor rise with temperature?

(iii) State Kirchhoff's second rule and write its mathematical formula.

(iv) Write any two properties of parallel resonance circuit.

(v) How does doubling the frequency affect the reactance (a) an inductor (b) a capacitor?

(vi) How the reception of a particular radio station is selected on your radio set?

(vii) Define modulus of elasticity. Show that units of modulus of elasticity and stress are the same.

(viii) What is meant by strain energy?

(ix) Distinguish between crystalline solids and amorphous solids.

(x) Why is the base current in a transistor very small?

(xi) Why charge carriers are not present in the depletion region?

(xii) Write the names of three parts of a transistor.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Does the induced emf in a circuit depend on the resistance of the circuit? Does the induced current depend on the resistance of the circuit?

(ii) Four unmarked wires emerge from a transformer. What steps would you take to determine the turns ratio?

(iii) What factors can change the mutual inductance of two coils?

(iv) If the length of solenoid is doubled keeping all other factors same, what will be change in energy density of current carrying solenoid?

(v) Why do not we observe a Compton's effect with visible light?

(vi) Will higher frequency light eject greater number of electrons than low frequency light?

(vii) Explain NAVSTAR navigation system.

(viii) Can the electron in the ground state of hydrogen absorb a photon of energy 13.6eV and greater than 13.6eV?

(ix) Explain bremsstrahlung in x-rays spectrum.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 05) State Gauss's law and apply it to find out the electric intensity due to infinite sheet of charge.

Q5(b). (Marks 03) A rectangular bar of iron is 2cm by 2cm in cross-section and 40cm long. Calculate its resistance. (Resistivity = $11 \times 10^{-8} \Omega m$)

Q6(a). (Marks 05) State Ampere's law and apply it to find the magnetic field due to a current carrying solenoid.

Q6(b). (Marks 03) When current through a coil changes from 100mA to 200mA in 0.005s, an induced emf of 40mV is produced in the coil. What is the self-inductance of the coil?

Q7(a). (Marks 05) What is transistor? Describe its construction and operation. Also show that how current flows in n p n transistor?

Q7(b). (Marks 03) Find the value of current flowing through a capacitance 0.5μF, when connected to a source of 150V at 50Hz.

Q8(a). (Marks 05) What are intrinsic and extrinsic semi-conductors? How the p-type and n-type materials are formed?

Q8(b). (Marks 03) X-rays of wavelength 22pm are scattered from a carbon target. The scattered radiation being viewed at 85° to the incident beam. What is Compton shift?

Q9(a). (Marks 05) Describe atomic spectrum of hydrogen. Show that energy of electron in hydrogen atom is quantized.

Q9(b). (Marks 03) Calculate the energy (in MeV) released in the following fusion reaction: ${}^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + {}^{1}_{0}n$.